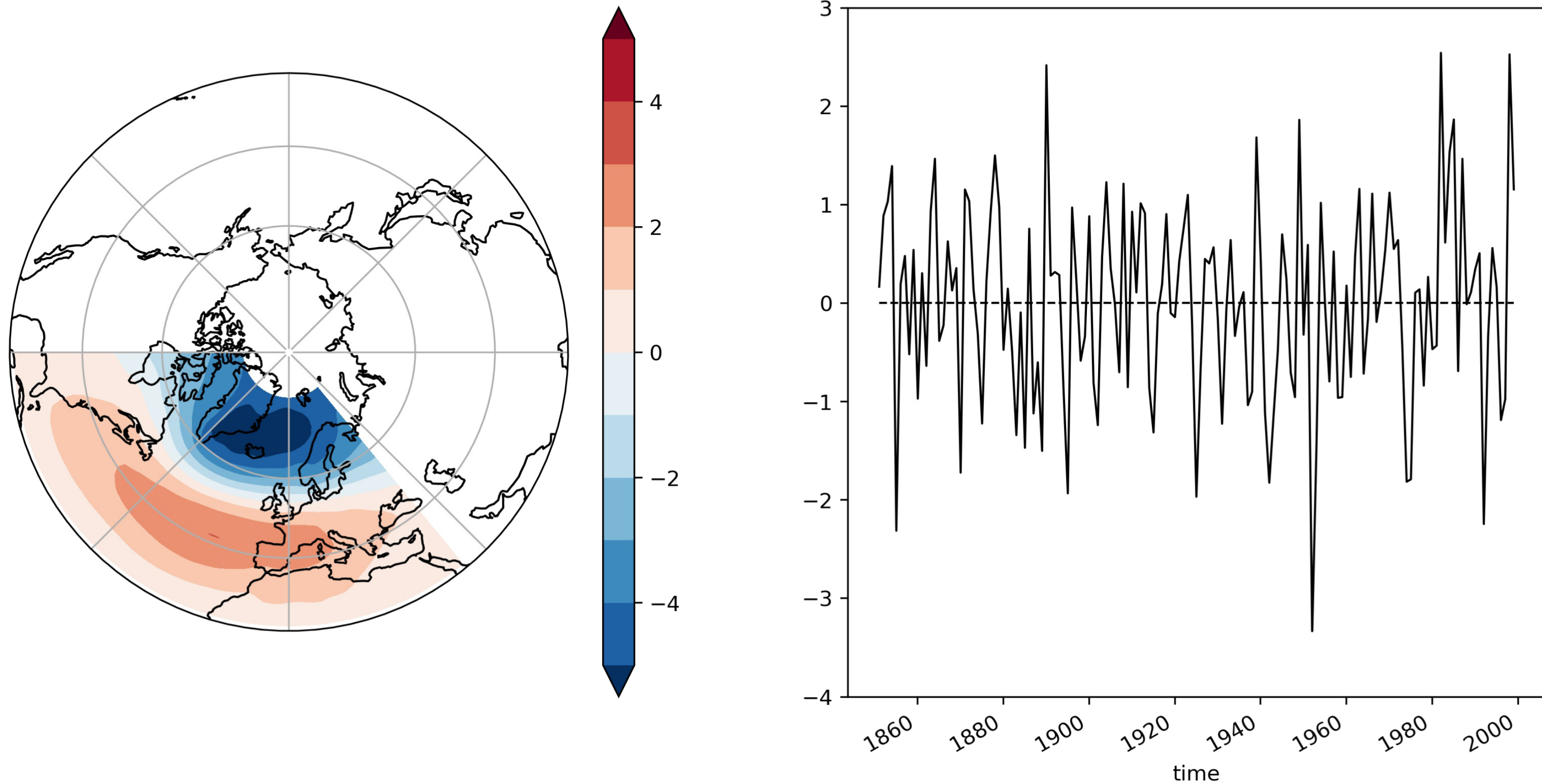


Predicting daily NAO index with seq2seq model

North Atlantic Oscillation (NAO)

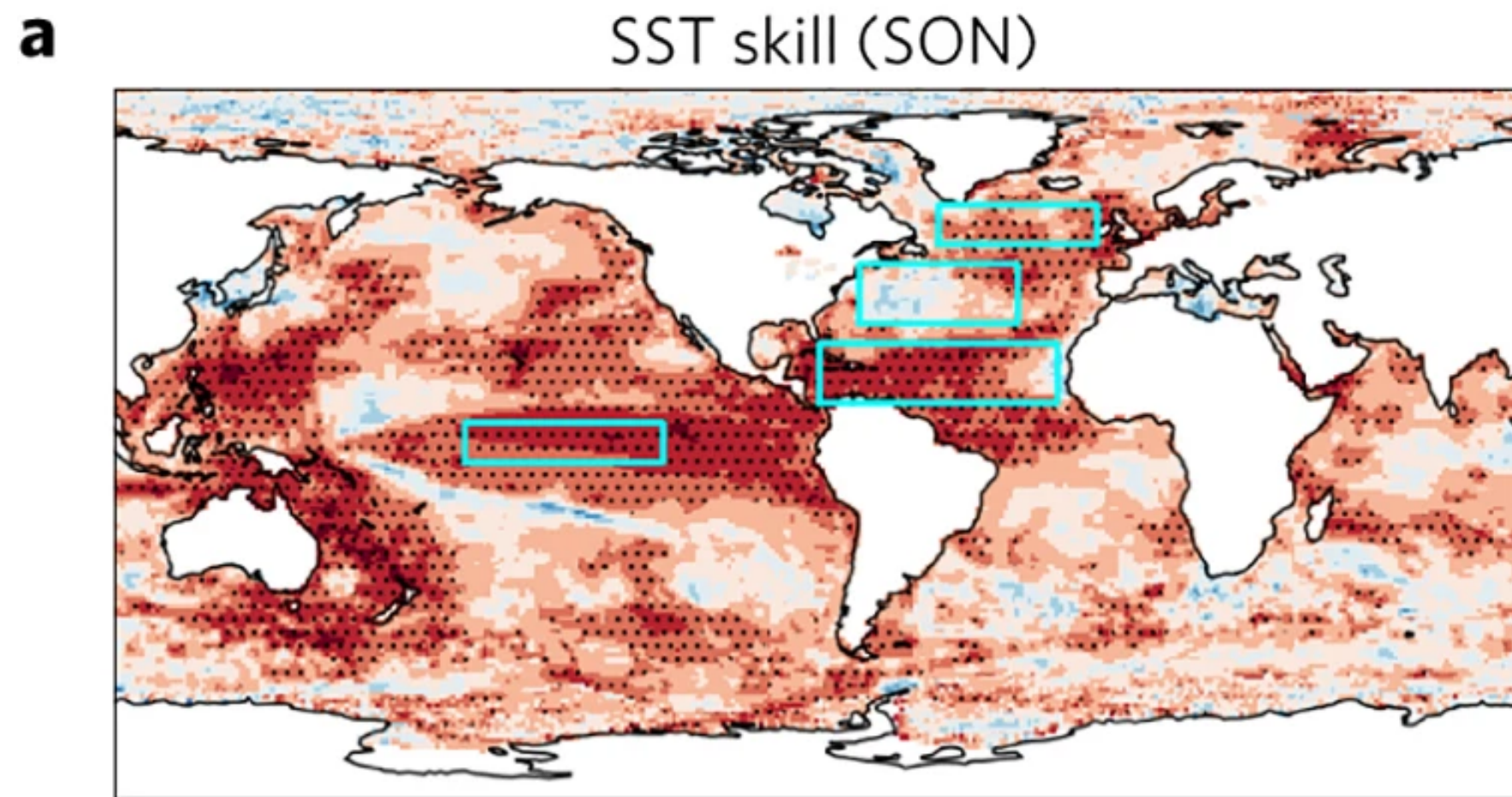
Yearly index



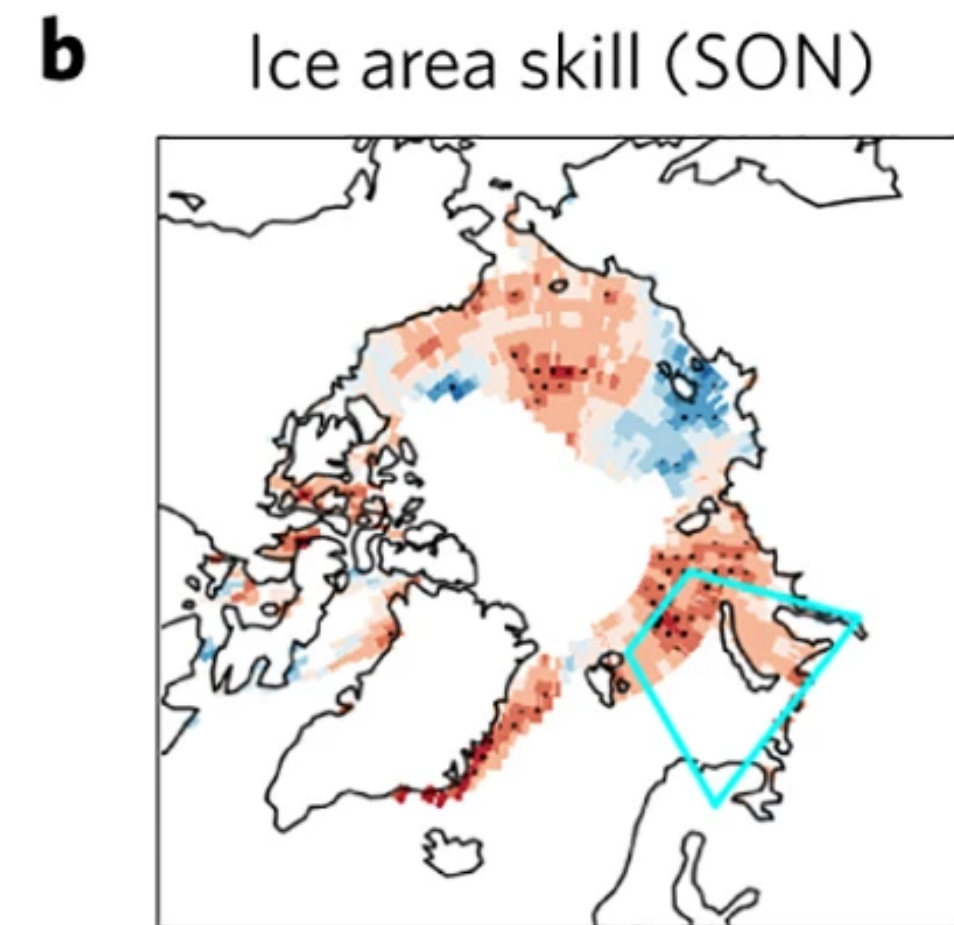
largely driven by chaotic and inherently unpredictable process

Predicting NAO one year lead

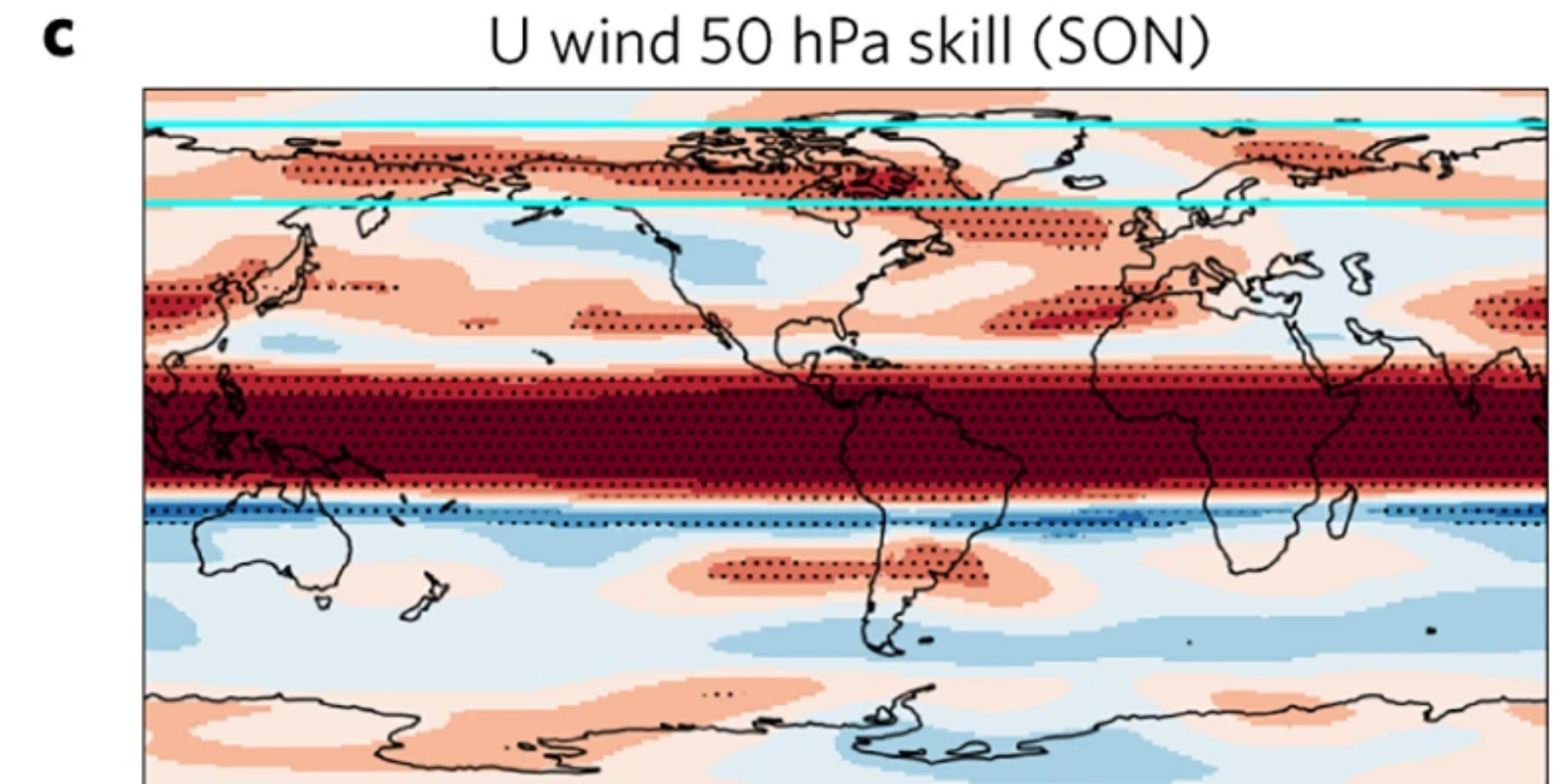
Seasonal index



ENSO: pacific Sea Surface Temperature
AST: Atlantic SST tripole pattern (AST)



SIC: sea ice coverage



SPVS: stratospheric polar vortex strength

Daily index of NAO from MPI-GE

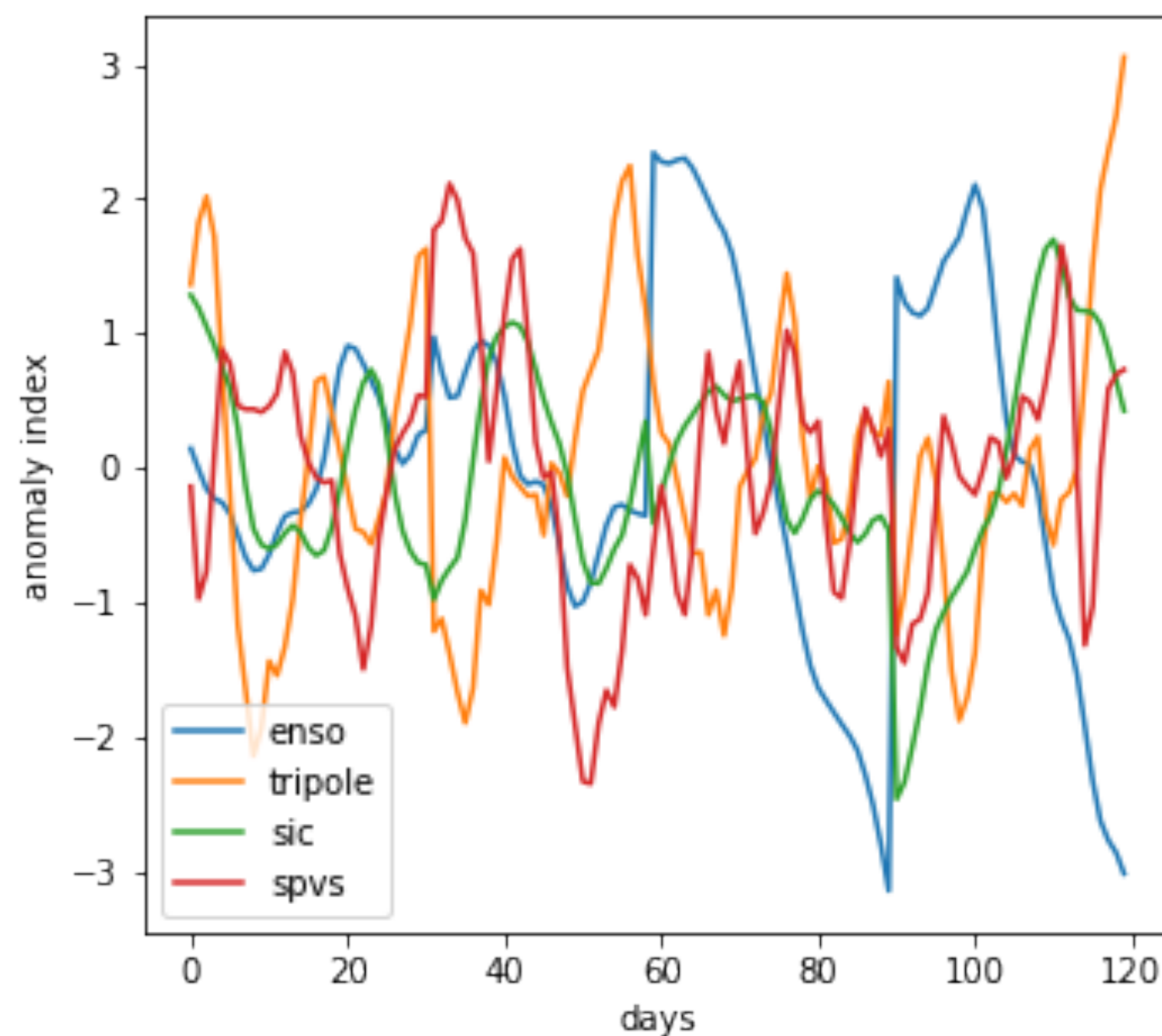
MPI-GE:

Historical runs: 100 ensembles

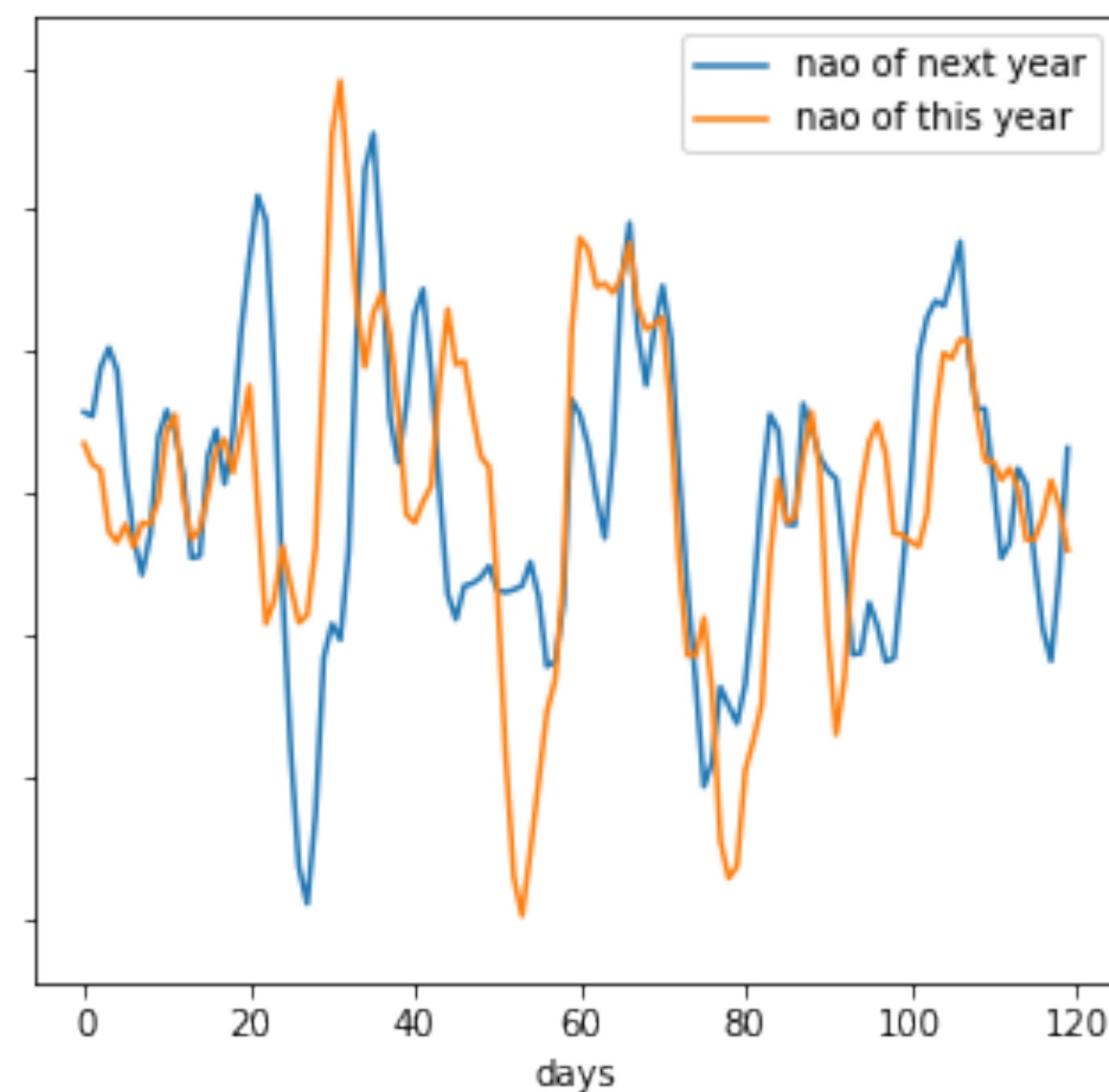
Training dataset: $10780 \times 120 \times 6$

Valid dataset: $3080 \times 120 \times 6$

Test dataset: $1540 \times 120 \times 6$

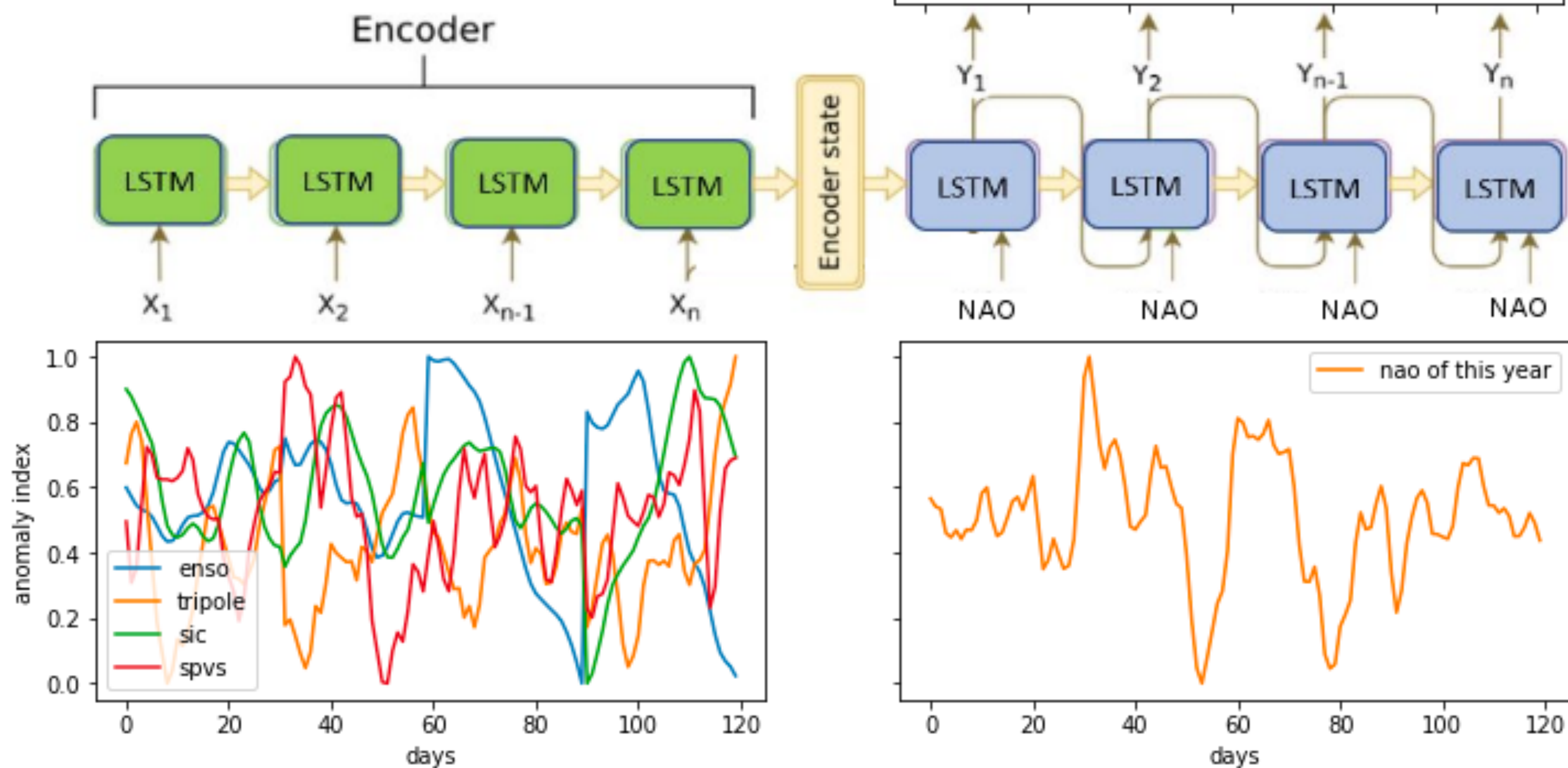


Input

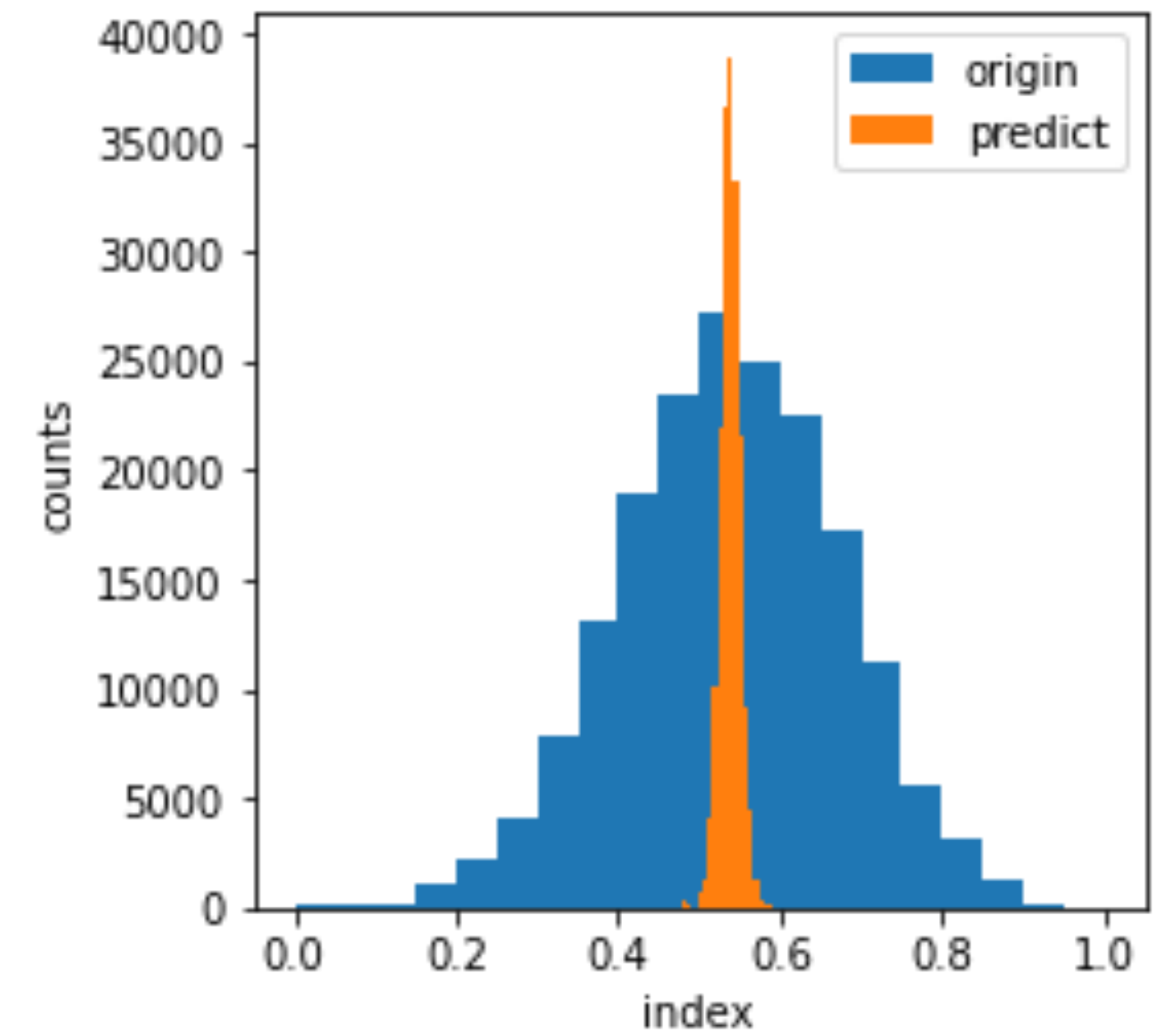
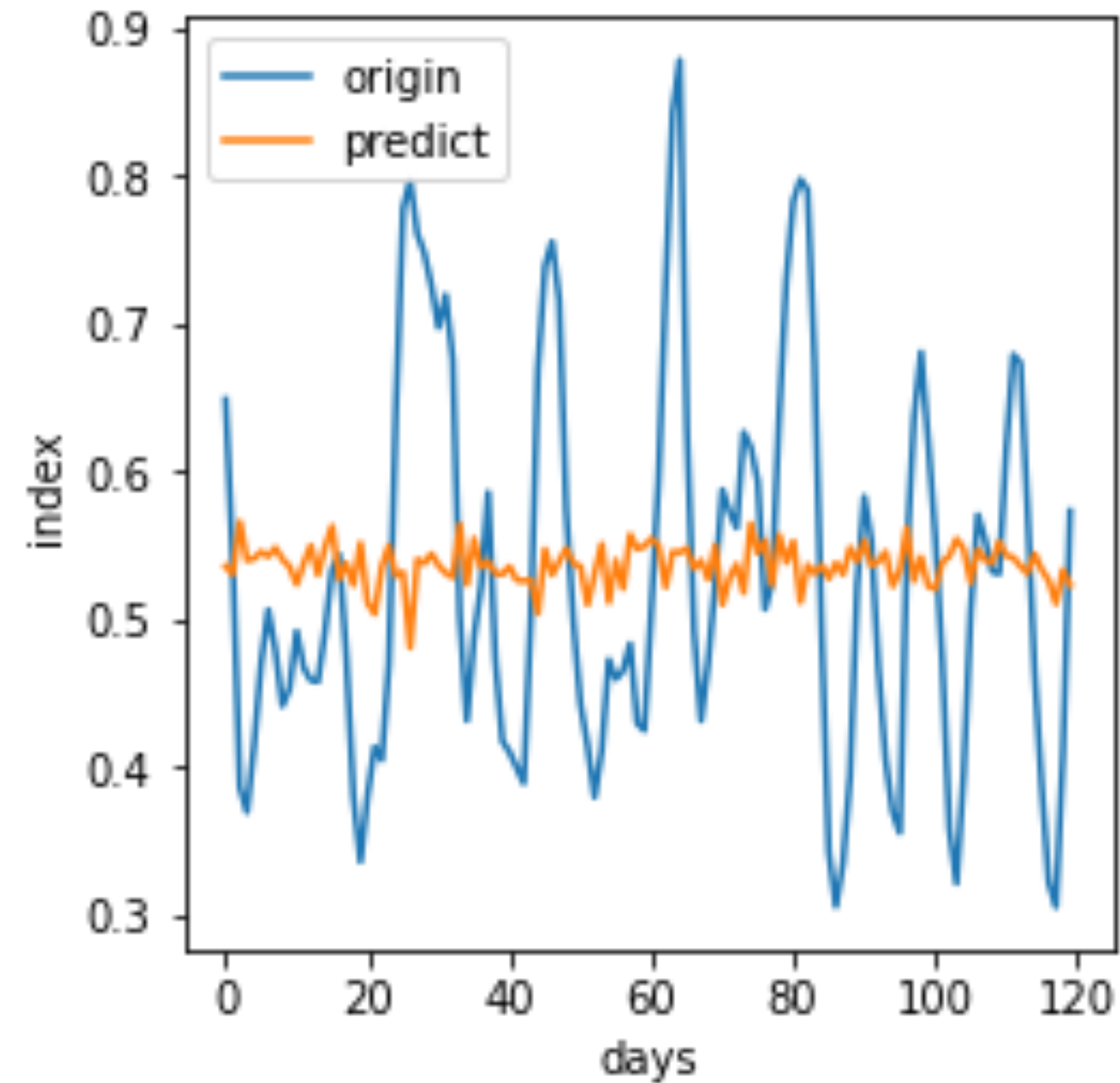
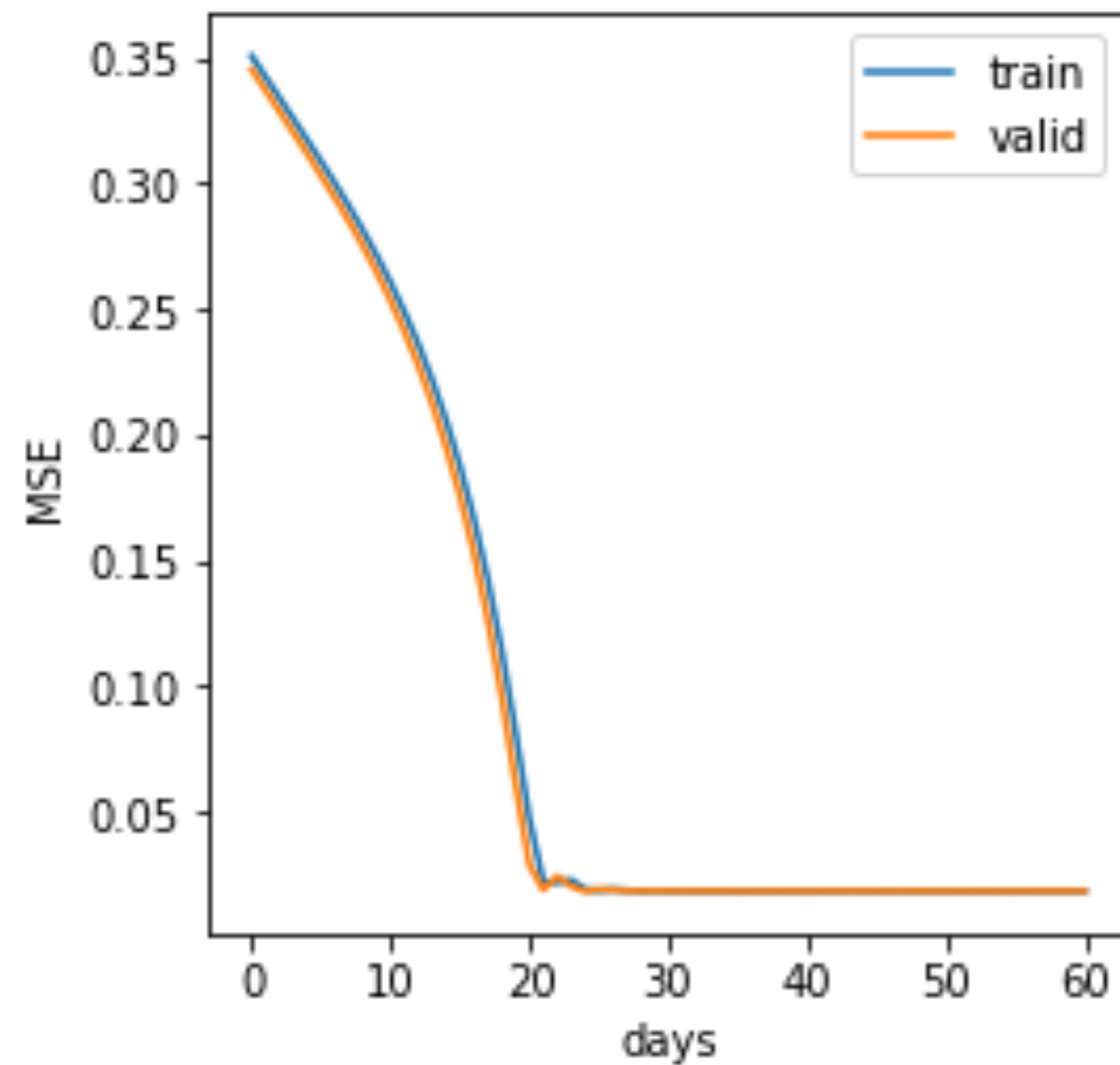


Output

Seq2seq model



Training and test



Reasonable absolute value, but largely underestimate the temporal variability

Experiment 2: change loss function

Include the temporal variability in loss function

Loss before:

$$MSE = (x - y)^2$$

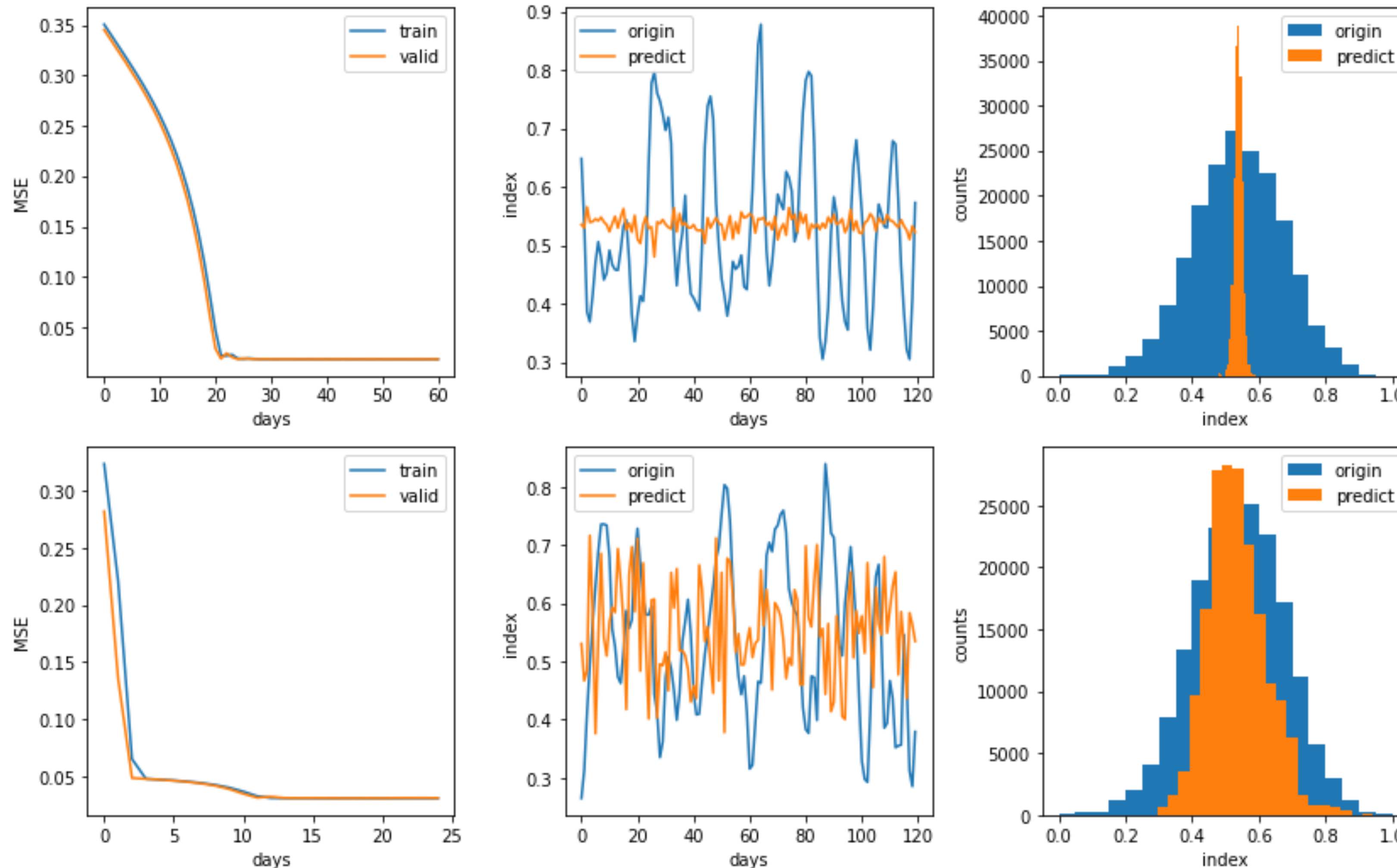
Loss now:

$$Loss = (x - y)^2 + \beta((x - \bar{x})^2 - (y - \bar{y}))^2$$

Difference of Standard deviation

Exp 2: change loss function

Include the temporal variability in loss function

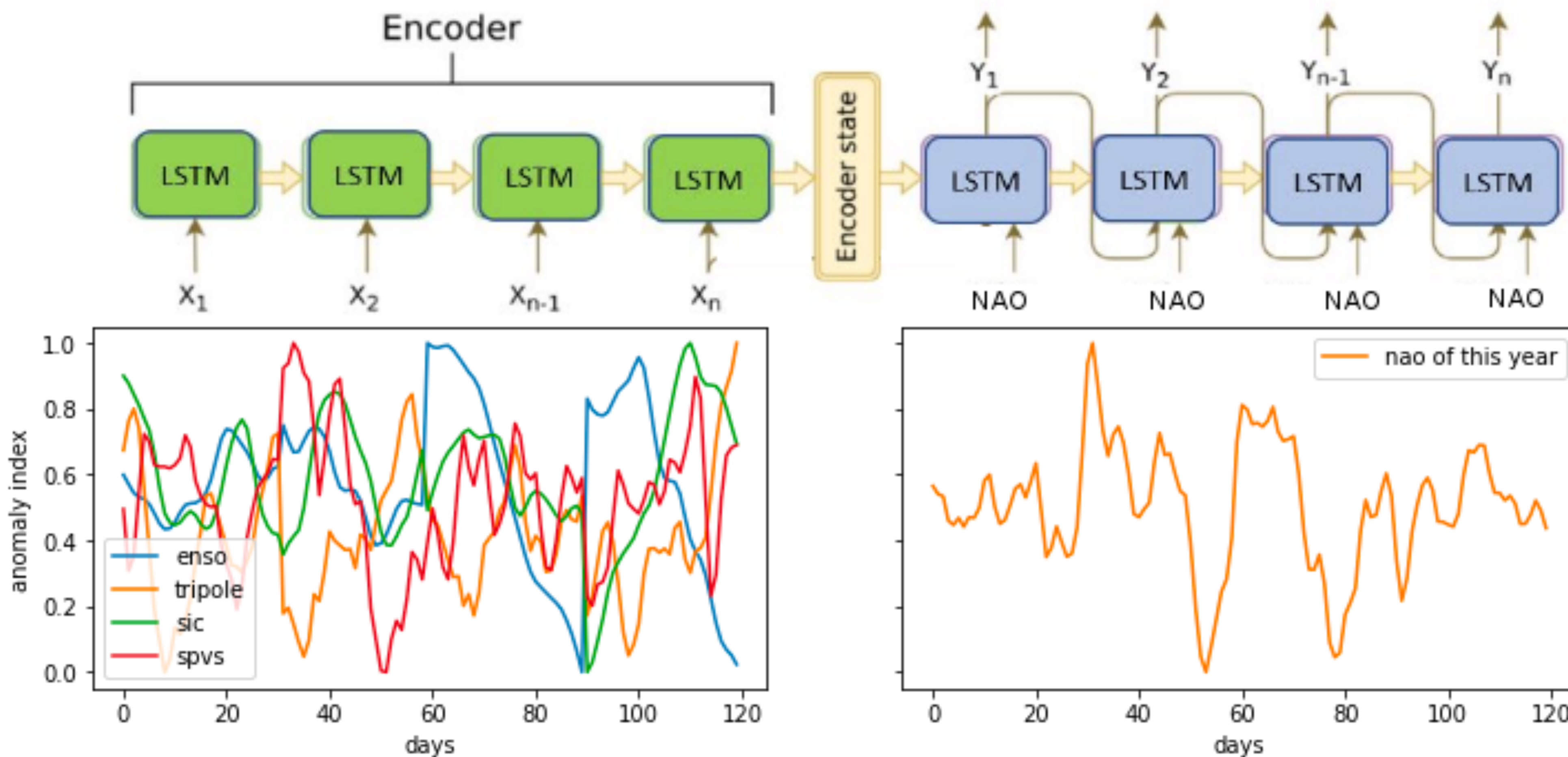


MSE only

MSE + temporal var
 $\beta = 2$

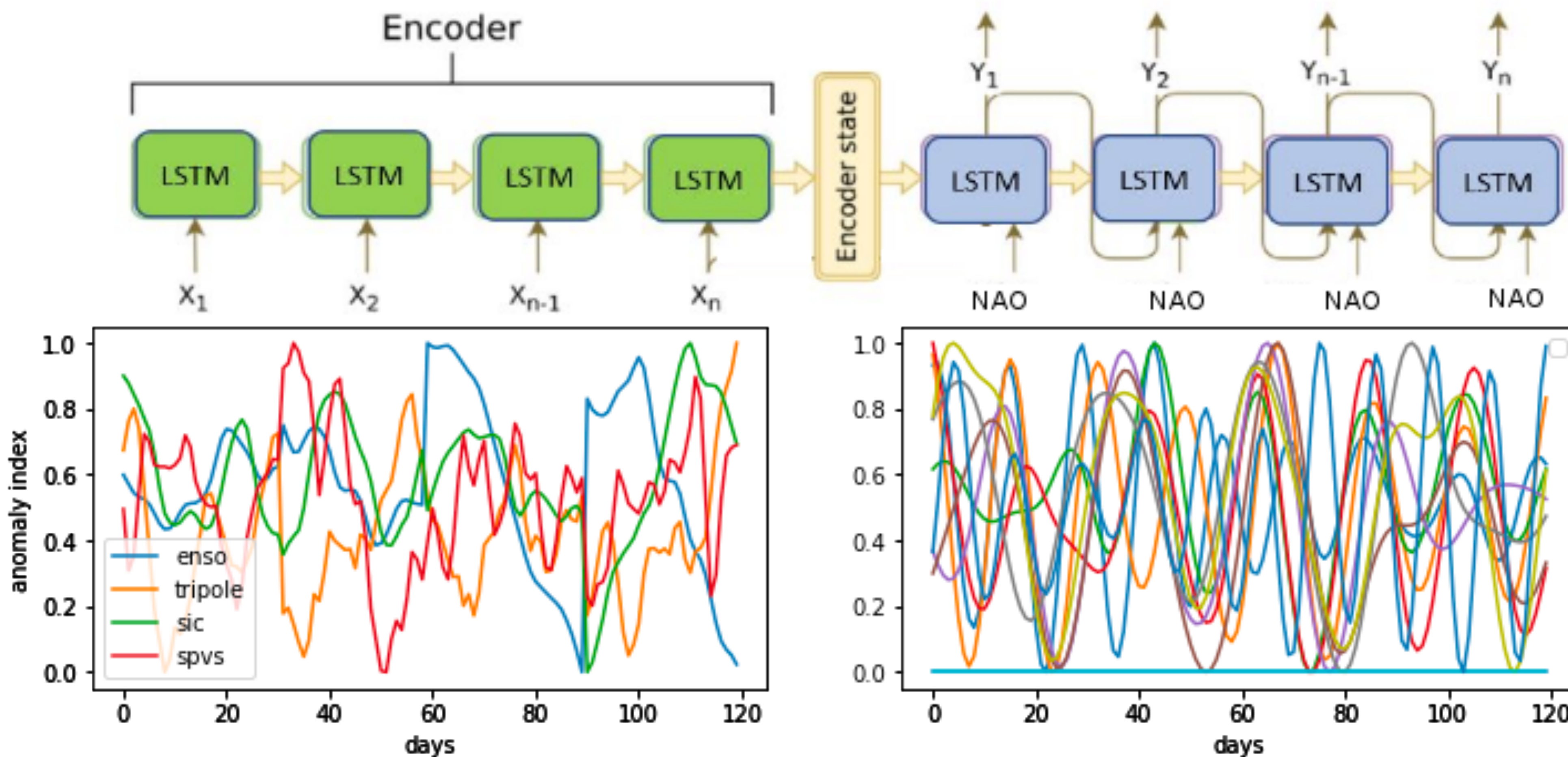
Exp3: add spectrum info

Include SSA components in decoder

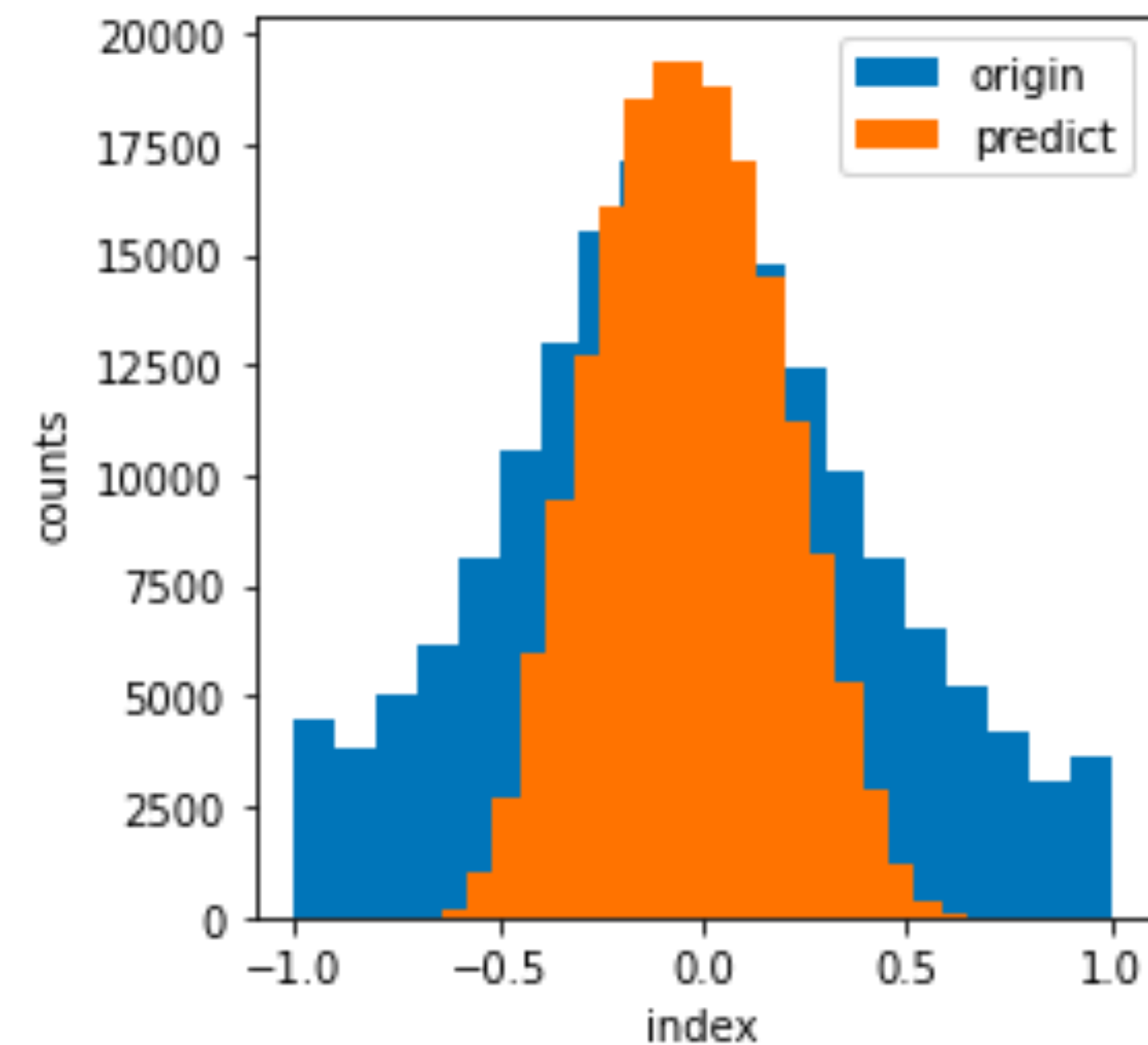
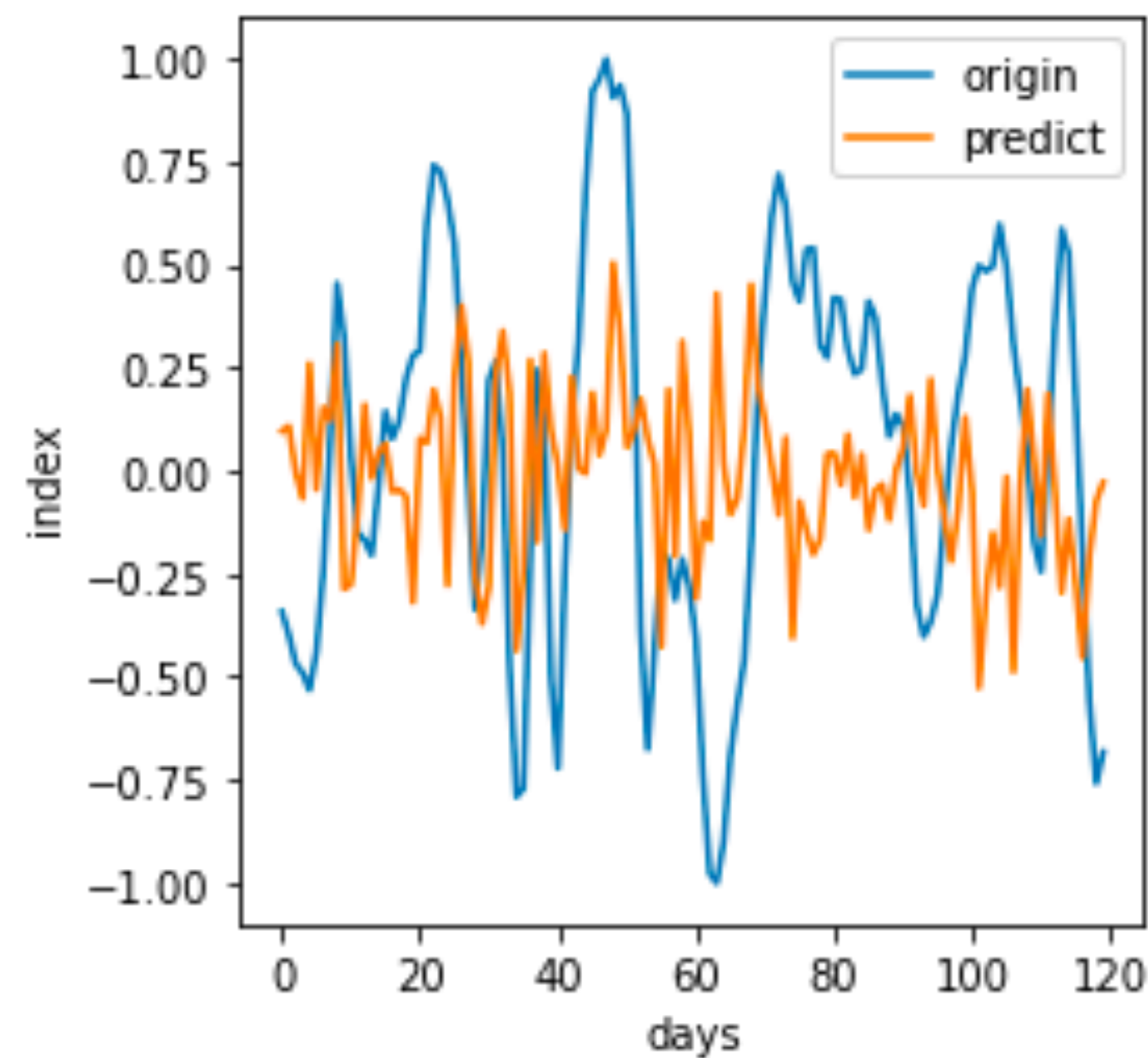
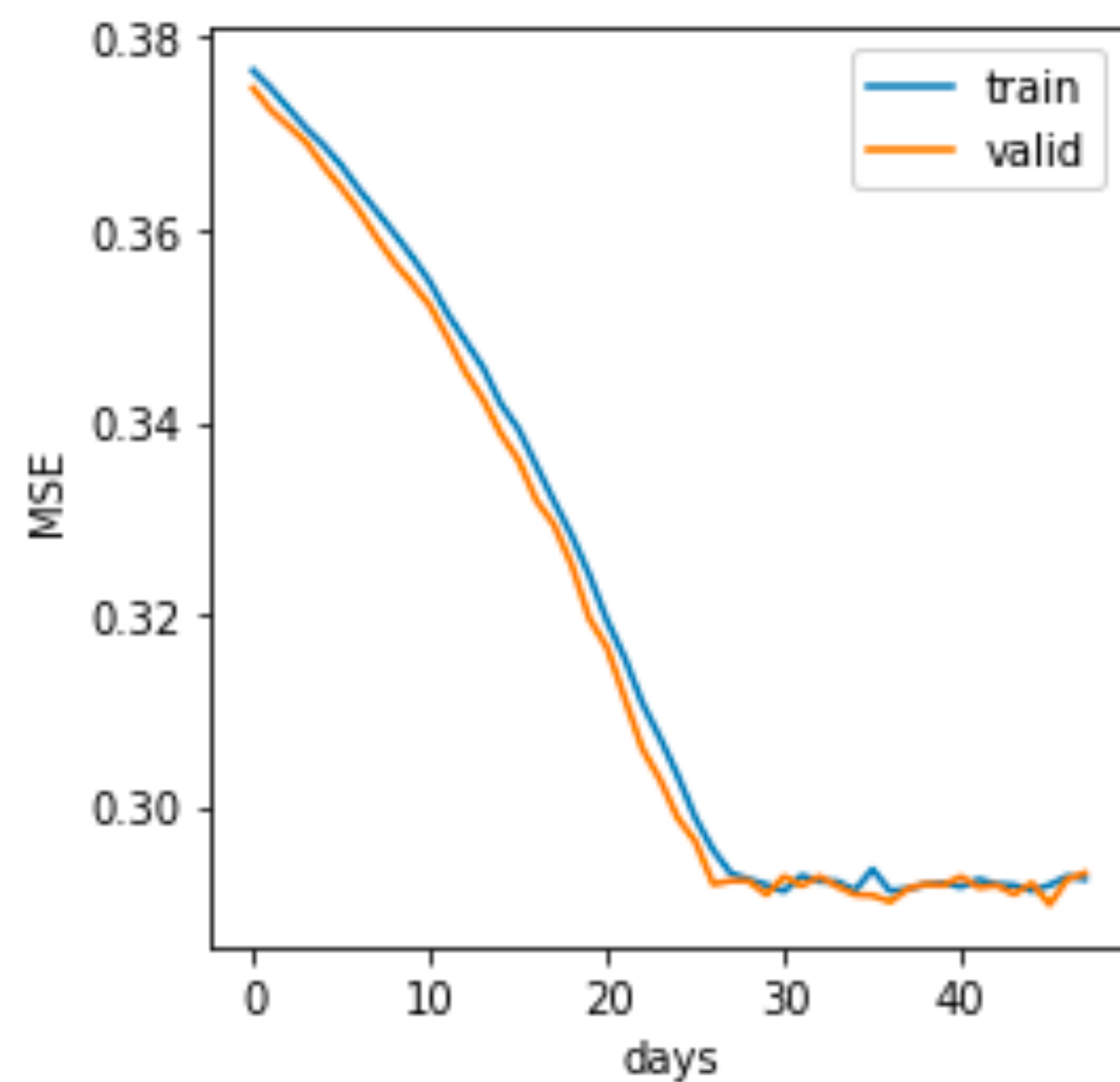


Exp3: add spectrum info

Include SSA components in decoder



Exp3: add spectrum info



Conclusion

1. Predicting the NAO index from other climate variables is a quite challenging task.
2. Seq2seq model can predict the seasonal average absolute value, but largely underestimated the temporal variability.
3. Accuracy could be partly improved by including temporal variabilities in loss function and adding spectrum information to decoder.